

Accounting Business and Society

–Inventory and Cost of Goods Sold

WEEK 6

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Key concepts

- 01** Recording, expensing & Reporting Inventories
- 02** Periodic Inventory system
- 03** Effects of Inventory Errors
- 04** Inventory system for COGS: FIFO, LIFO and Average Costing

1#Recording, expensing & Reporting Inventories

- The cost of any asset, such as inventory, is the sum of all the costs incurred to bring the asset to its intended use, less any discounts. Cost includes:
 - Purchase price
 - Freight in
 - Insurance while in transit
 - Fees/taxes to get inventory ready to sell
 - Less returns, allowances, and discounts

1# Recording for Inventory in the Perpetual System

Purchases of Goods on credit

Date	Accounts	Debit	Credit
	Inventory	\$\$\$	
	Accounts Payable		\$\$\$
	To record acquisition of inventory		

1# Recording for Inventory in the Perpetual System

	Debit	Credit
Purchase returns:		
Accounts payable	500	
Inventory		500
<i>To record purchase return of merchandise that cost \$500.</i>		
Purchase discounts:		
Original purchase:		
Inventory (1,000 × 1)	1,000	
Accounts payable (1,000 × 1)		1,000
<i>To record a gross purchase of \$1,000 in merchandise.</i>		
Payment 10 days later (within discount period):		
Accounts payable	1,000	
Inventory		20
Cash		980
<i>To record payment for merchandise within 10 days at 2% discount.</i>		

1# Recording for Inventory in the Perpetual System

The cost of inventory is the net amount of purchases, determined as follows (using assumed amounts):

Purchase price of the inventory	\$600,000
+ Freight in (the cost to transport the goods from the seller to the buyer)	4,000
– Purchase returns for unsuitable goods returned to the seller.....	(25,000)
– Purchase allowances granted by the seller	(5,000)
– Purchase discounts for early payment by the buyer	<u>(14,000)</u>
= Net purchases of inventory—Cost to the buyer	<u><u>\$560,000</u></u>

1#Recording, Expensing & Reporting Inventories

The cost of inventory sold shifts from asset to expense when the seller delivers goods to the buyer.

The cost
of inventory
on hand
= Inventory

**Asset on the
Balance Sheet**

The cost
of inventory
that's been sold
= Cost of Goods Sold

**Expense on the
Income Statement**

1#Recording, Expensing & Reporting Inventories

Sales of Goods & **Expensing** Inventory

4				
5	2.	Accounts Receivable	900,000	
6		Sales Revenue		900,000
7		<i>Sold inventory on account.</i>		
8		Cost of Goods Sold	540,000	
9		Inventory		540,000
10		<i>Recorded cost of goods sold.</i>		
11				

1#Recording, expensing & Reporting Inventories

A1			
	A	B	C
1	Under Armour, Inc.		
2	Consolidated Balance Sheet (Partial, Adapted)		
3	December 31, 2014		
4	Assets		
5	(in millions)		
6	Current assets		
7	Cash and cash equivalents		\$ 593.2
8	Accounts receivable, net		279.8
9	Inventories		536.7

A1			
	A	B	C
1	Under Armour, Inc.		
2	Consolidated Statement of Income (Partial, Adapted)		
3	Year Ended December 31, 2014		
4	(In millions)		
5	Net (sales) revenues		\$ 3,084.4
6	Cost of goods sold		1,572.2
7	Gross profit (margin)		\$ 1,512.2

1#Recording, expensing & Reporting Inventories

- **Sales Price vs. Cost of Inventory**

- **Sales revenue** is based on the sale price of inventory sold
- **Cost of goods sold** is based on the cost of inventory sold

Cost of goods sold (income statement) = Number of units of inventory sold 證 ost per unit of inventory

- **Inventory** on the balance sheet is based on the cost of inventory still on hand

Inventory (balance sheet) = Number of units of inventory on hand 證 ost per unit of inventory

1#Recording, expensing & Reporting Inventories

Assume Under Armour Outlet, has in stock 300 men's fleece hoodies that cost \$30 each. The store marks each hoodie up by \$20 and sells 200 of the hoodies for \$50 each.

Balance Sheet (partial)

Current assets

Cash.....	\$XXX
Accounts receivable.....	XXX
Inventory (100 hoodies @ \$30 each)	3,000
Prepaid expenses	XXX

Income Statement (partial)

Sales revenue

(200 hoodies @ \$50 each)	\$10,000
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Cost of goods sold

(200 hoodies @ \$30 each)	<u>6,000</u>
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Gross profit.....	\$ 4,000
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1#Recording, expensing & Reporting Inventories

- Number of Units of Inventory
 - Determined from accounting records
 - Evidenced by physical count at year-end
 - Consigned goods:
 - Do **not** include those held for another company
 - Include those out on consignment
 - In transit goods treatment depends on shipping terms

1#Recording, expensing & Reporting Inventories

Number of Units of Inventory

- Company with legal title to the goods while in transit pays the transportation costs.
 - FOB Shipping Point
 - Ownership changes hands when the goods leave the seller's shipping dock
 - FOB Destination
 - Ownership changes hands when the goods are delivered to the customer



1#Recording, expensing & Reporting Inventories

- **Cost per Unit Inventory**
 - Poses challenges due to differing prices throughout the year



2# Perpetual and Periodic System for COGS

Perpetual Inventory System

- Used for all types of goods
- Keeps a running record of all goods bought, sold, and on hand
- Inventory counted at least once a year

Periodic Inventory System

- Used for inexpensive goods
- Does **not** keep a running record of all goods bought, sold, and on hand
- Inventory counted at least once a year

2# Perpetual and Periodic System for COGS

- **How the Perpetual System Works**
 - Clerk scans bar code of items you buy, optical scanner reads the bar code and:
 - Records sale
 - Updates inventory records
 - Two entries needed for each sale
 - Record revenue and asset received
 - Record cost of goods sold and reduction of inventory



Exhibit 6-4 Bar Code for Electronic Scanner

Exhibit 6-5 Recording and Reporting Inventory— Perpetual System (1 of 3)

A1				
	A	C	D	G
1	1.	Inventory	560,000	
2		Accounts Payable		560,000
3		<i>Purchased inventory on account.</i>		
4				
5	2.	Accounts Receivable	900,000	
6		Sales Revenue		900,000
7		<i>Sold inventory on account.</i>		
8		Cost of Goods Sold	540,000	
9		Inventory		540,000
10		<i>Recorded cost of goods sold.</i>		
11				

2# Periodic Inventory System

❑ Inventory & COGS will be updated at the end of period, not during the period.

❑ Temporary accounts used for recording:

- Purchase
- Transportation-in
- Purchase return and allowance
- Purchase Discount

❑ These accounts will be closed to Inventory at the end of period.

➤ **Net purchase=**

Purchase + Transportation-in – Purchase return & allowance – Purchase discount



2# Periodic Inventory System

❑ Inventory is **never used** for recording during the period.

Purchase of Goods

Date	Accounts	Debit	Credit
xxx	Purchase	xxx	
	Account Payable		xxx

Transportation-in

Date	Accounts	Debit	Credit
xxx	Purchase or (Transportation-in)	xxx	
	Cash/ Account payable		xxx

Purchase return
&
Allowance

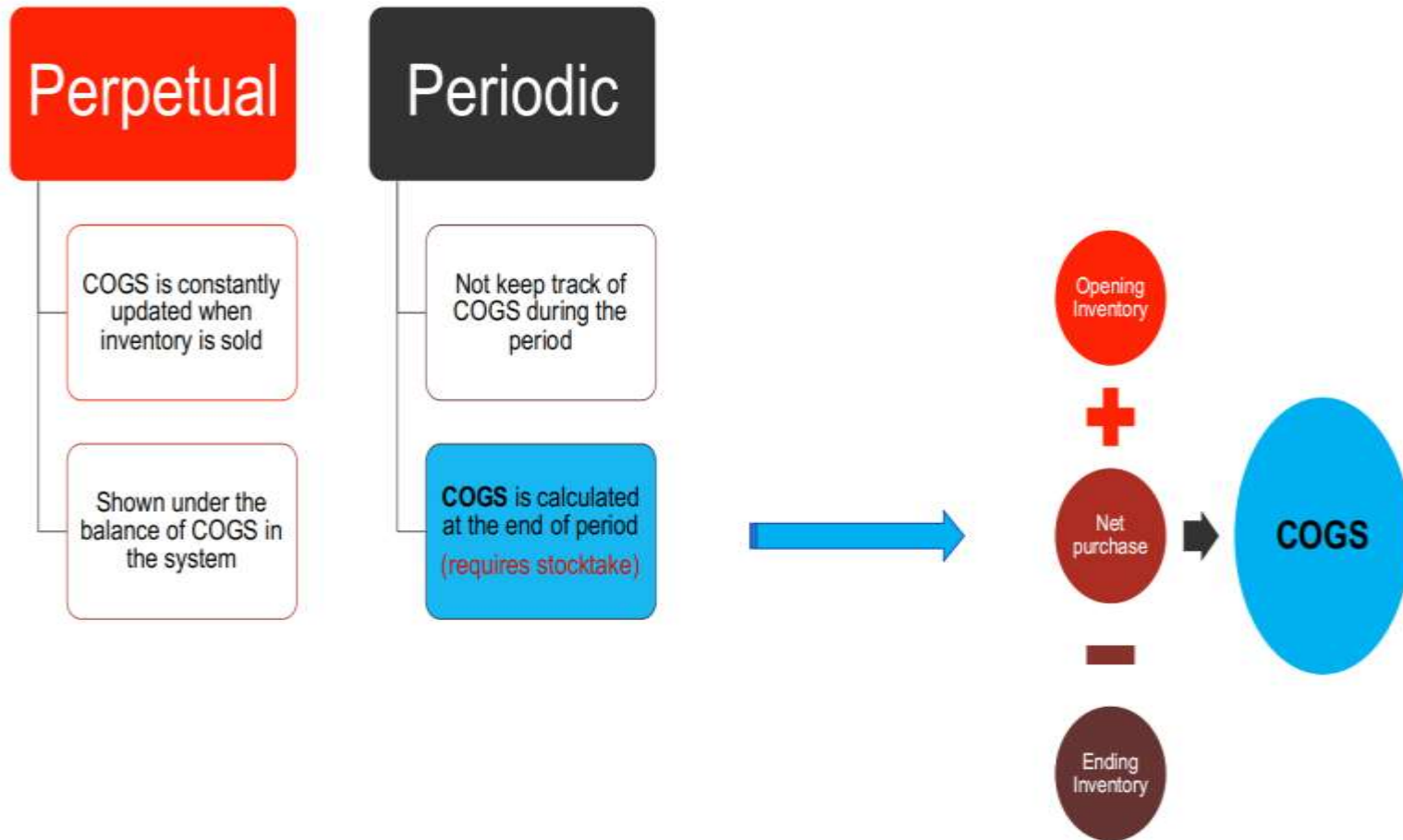
Date	Accounts	Debit	Credit
xxx	Account payable	xxx	
	Purchase return & allowance		xxx

Purchase Discount

Date	Accounts	Debit	Credit
xxx	Account payable	xxx	
	Purchase Discount		xxx

❑ **Note:** This is the use of **Periodic Inventory System**

2# Periodic System for COGS



3# Analyze Effects of Inventory Errors

- Error in ending inventory creates errors for two accounting periods
- Inventory errors counterbalance in two consecutive periods
- Beginning inventory and ending inventory have opposite effects on cost of goods sold
 - Beginning inventory is added
 - Ending inventory is subtracted

Exhibit 6-14 Inventory Errors: An Example

A1							
	A	B	C	D	E	F	G
1		Period 1		Period 2		Period 3	
2		Ending Inventory		Beginning Inventory		Correct	
3		Overstated by \$10,000		Overstated by \$10,000			
4						Correct	
5	Sales revenue		\$ 100,000		\$100,000		\$ 100,000
6	Cost of goods sold:						
7	Beginning inventory	\$ 10,000		\$ 20,000		\$ 10,000	
8	Purchases	50,000		50,000		50,000	
9	Cost of goods available	60,000		70,000		60,000	
10	Ending inventory	(20,000)		(10,000)		(10,000)	
11	Cost of goods sold		40,000		60,000		50,000
12	Gross profit		\$ 60,000		\$ 40,000		\$ 50,000
13							
14				\$ 100,000			
15							
16							

Exhibit 6-15 Effects of Inventory Errors

A1					
	A	B	C	D	E
		Period 1		Period 2	
		Cost of Goods Sold	Gross Profit and Net Income	Cost of Goods Sold	Gross Profit and Net Income
1					
2	Period 1				
3	Ending inventory overstated	Understated	Overstated	Overstated	Understated
4	Period 1				
5	Ending inventory understated	Overstated	Understated	Understated	Overstated
6					

4# Inventory system for COGS: FIFO, LIFO and Average Costing

- The accounting method selected by management affects the:
 - Profits reported
 - Income tax paid
 - Values of inventory turnover and gross margin percentage ratios

Exhibit 6-6 Inventory Data Used to Illustrate the Various Inventory Costing Methods

Inventory			
Beg bal	(10 units @ \$10)	100	
Purchases:			Cost of goods sold
No. 1	(25 units @ \$14)	350	(40 units @ ?)
No. 2	(25 units @ \$18)	450	
End bal	(20 units @ ?)	?	

Goods Available		Number of Units	Total Cost
Goods available	=	10 + 25 + 25 = 60 units	\$100 + \$350 + \$450 = \$900
Cost of goods sold	=	40 units	?
Ending inventory	=	20 units	?

4# Inventory System for COGS

- Accounting uses four generally accepted inventory methods:
 - Specific unit cost
 - Average cost
 - First-in, first-out (FIFO) cost
 - Last-in, first-out (LIFO) cost



4# Inventory System for COGS

- **Specific unit cost**

- Used for businesses with unique inventory items
- Automobiles, antique furniture, & real estate
- Businesses cost their inventories at the specific cost of the particular unit
- Too expensive for inventories with common characteristics
- Also called the specific identification method

4# Inventory System for COGS

- **Average Cost**
 - Also called the weighted-average method
 - Based on the average cost of inventory during the period

Average Cost Method (1 of 2)

Inventory			
Beg bal	(10 units @ \$10)	100	
Purchases:			Cost of goods sold
No. 1	(25 units @ \$14)	350	(40 units @ ?)
No. 2	(25 units @ \$18)	450	
End bal	(20 units @ ?)	?	

$$\text{Average cost per unit} = \frac{\text{Cost of goods available}^1}{\text{Number of units available}} = \frac{\$900}{60} = \$15$$

$$\begin{aligned} \text{Cost of goods sold} &= \text{Number of units sold} \times \text{Average cost per unit} \\ &= 40 \text{ units} \times \$15 = \$600 \end{aligned}$$

$$\begin{aligned} \text{Ending inventory} &= \text{Number of units on hand} \times \text{Average cost per unit} \\ &= 20 \text{ units} \times \$15 = \$300 \end{aligned}$$

Average Cost Method (2 of 2)

Inventory (at Average Cost)			
Beg bal	(10 units @ \$10)	100	
Purchases:			
No. 1	(25 units @ \$14)	350	
No. 2	(25 units @ \$18)	450	
			Cost of goods sold (40 units @ average cost of \$15 per unit) 600
End bal	(20 units @ average cost of \$15 per unit)	300	

4# Inventory System for COGS

- **First-in, first out (FIFO)**
 - First costs into inventory are the first costs assigned to cost of goods sold
 - Ending inventory is based on the latest costs incurred

FIFO Cost

Inventory (at FIFO cost)				
Beg bal	(10 units @ \$10)	100		
Purchases:			Cost of goods sold (40 units):	
No. 1	(25 units @ \$14)	350	(10 units @ \$10)	100
No. 2	(25 units @ \$18)	450	(25 units @ \$14)	350
			(5 units @ \$18)	90
End bal	(20 units @ \$18)	360	} 540	

4# Inventory System for COGS

- **Last-in, first-out (LIFO)**
 - Costing is the opposite of FIFO
 - Last costs into inventory go immediately to cost of goods sold
 - Ending inventory is based on the oldest costs

LIFO Cost

Inventory (at LIFO cost)					
Beg bal	(10 units @ \$10)	100			
Purchases:			Cost of goods sold (40 units):		
No. 1	(25 units @ \$14)	350	(25 units @ \$18)	450	} 660
No. 2	(25 units @ \$18)	450	(15 units @ \$14)	210	
End bal	(10 units @ \$10)	} 240			
	(10 units @ \$14)				

Exhibit 6-7 Income Effects of the FIFO, LIFO, and Average Inventory Methods; Increasing Costs

	FIFO	LIFO	Average
Sales revenue (assumed)	\$1,000	\$1,000	\$1,000
Cost of goods sold.....	<u>540 (lowest)</u>	<u>660 (highest)</u>	<u>600</u>
Gross profit.....	<u><u>\$ 460 (highest)</u></u>	<u><u>\$ 340 (lowest)</u></u>	<u><u>\$ 400</u></u>

Exhibit 6-8 Cost of Goods Sold and Ending Inventory— FIFO and LIFO; Increasing Costs and Decreasing Costs (1 of 2)

Panel A — When Inventory Costs Are Increasing

	Cost of Goods Sold (COGS)	Ending Inventory (EI)
FIFO	FIFO COGS is lowest because it's based on the oldest costs, which are low. Gross profit is, therefore, the highest.	FIFO EI is highest because it's based on the most recent costs, which are high.
LIFO	LIFO COGS is highest because it's based on the most recent costs, which are high. Gross profit is, therefore, the lowest.	LIFO EI is lowest because it's based on the oldest costs, which are low.

Exhibit 6-8 Cost of Goods Sold and Ending Inventory—FIFO and LIFO; Increasing Costs and Decreasing Costs (2 of 2)

Panel B — When Inventory Costs Are Decreasing

	Cost of Goods Sold (COGS)	Ending Inventory (EI)
FIFO	FIFO COGS is higher because it's based on the oldest costs, which are high. Gross profit is, therefore, the lowest.	FIFO EI is lowest because it's based on the most recent costs, which are low.
LIFO	LIFO COGS is lowest because it's based on the most recent costs, which are low. Gross profit is, therefore, the highest.	LIFO EI is highest because it's based on the oldest costs, which are high.

4# Inventory System for COGS

- **Tax Advantages of LIFO**

- When costs are rising → LIFO results in lowest taxable income → which results in the lowest income taxes → increasing cash that is available

	FIFO	LIFO
Gross profit (from Exhibit 6-7)	\$460	\$340
Operating expenses (assumed).....	<u>260</u>	<u>260</u>
Income before income tax	<u>\$200</u>	<u>\$ 80</u>
Income tax expense (40%)	<u>\$ 80</u>	<u>\$ 32</u>

4# Inventory System for COGS

- **Tax Advantages of LIFO**

- Measuring cost of goods sold
 - LIFO → more realistic net income figure
 - Most recent costs assigned to Cost of Goods Sold
- Measuring ending inventory
 - FIFO → provides more up-to-date inventory cost
 - More recent costs on the balance sheet

4# Inventory System for COGS

- **LIFO and Managing Reported Income**
 - LIFO allows for manipulation of net income
 - Timing of purchases of inventory
 - When inventory prices are rising, purchase large quantities near year end to pay lower taxes
 - LIFO Liquidation
 - Quantities fall below the level from the previous period
 - Must dip into older layers of inventory costs, shifting older, lower costs into cost of goods sold

Extention

Explain and apply underlying GAAP for inventory



Explain and Apply the Underlying GAAP for Inventory (1 of 4)

- **Disclosure**
 - Financial statements should report enough information for outsiders to make informed decisions
- **Representational Faithfulness**
 - Properly disclosing inventory methods and substance of all material transactions
- **Consistency**
 - Use comparable methods from period to period

Explain and Apply the Underlying GAAP for Inventory (2 of 4)

Lower-of-Cost-or-Market Rule

- Based on principles of relevance and representational faithfulness
- Require inventory be reported on the balance sheet at whichever is lower:
 - Inventory's historical cost, or
 - Market value
 - Generally means current replacement cost

Explain and Apply the Underlying GAAP for Inventory (3 of 4)

Lower-of-Cost-or-Market Rule. Under Armour, Inc., paid \$3,000,000 for inventory. By its year-end, the inventory can be replaced for \$2,000,000. Under Armour's year-end balance sheet must report this inventory at the LCM value of \$2,000,000.

A1				
	A	B	C	D
1	Dec. 31	Cost of Goods Sold	1,000,000	
2		Inventory		1,000,000
3		<i>Wrote inventory down to market value.</i>		
4				

Exhibit 6-9 Lower-of-Cost-or-Market (LCM) Effects on Inventory and Cost of Goods Sold

Balance Sheet

Current assets:

Cash	\$ XXX,XXX
Short-term investments	XXX,XXX
Accounts receivable.....	XXX,XXX
Inventories, at market	
(which is lower than \$3,000,000 cost)	2,000,000
Prepaid expenses	XXX,XXX
Total current assets	<u>\$X,XXX,XXX</u>

Income Statement

Sales revenue	\$21,000,000
Cost of goods sold (\$9,000,000 + \$1,000,000) ...	<u>10,000,000</u>
Gross profit.....	\$11,000,000

Assessment

Tutorial portfolio - Online quizzes:

This quiz is based on the week 6 learning materials. You may complete this anytime during week 6, you may use your study notes.

You have TWO attempts to complete this quiz.

Due time for your weekly quiz is on Canvas, please check it by yourself.